

K. SUCHIR REDDY

Software Developer Intern · Backend & Systems

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PROFESSIONAL SUMMARY

4th-year Computer Science undergraduate at DSCE (CPI: 8.15) with solid foundations in software systems, algorithms, and clean architecture in Python, C++, Java, and SQL. Practical experience building AST syntax analysis pipelines and modular REST APIs.

EDUCATION

Dayananda Sagar College of Engineering (DSCE), Bengaluru
Bachelor of Engineering (B.E.) — Computer Science & Business Systems

CPI: 8.15 / 10.0
4th Year Undergraduate (Batch 2026)

TECHNICAL SKILLS

Languages: Python (Advanced), Java, C++, C, SQL

Systems & Backend: Flask, FastAPI, RESTful APIs, MySQL, DBMS, AST Parsing

Tools & Frameworks: Git / GitHub, Docker (Basics), AWS S3, Linux, PyTorch

KEY ENGINEERING PROJECTS

Multi-Model Code Generation & Benchmarking Engine

[Python, PyTorch, Transformers, AST (Abstract Syntax Trees), Evaluation Metrics]

- Built an automated evaluation pipeline measuring LLM-generated code across runtime latency, memory, and syntactic correctness.
- Utilized Abstract Syntax Tree (AST) parsing in Python to validate grammatical code structures without execution overhead.
- Benchmarked multi-model inference performance across varied prompt complexities with structured JSON logging.

AI Content Moderator & Social Platform

[Python, Flask, FastAPI, NLP, MySQL, Scikit-learn, REST APIs]

- Engineered high-throughput REST APIs in Flask/FastAPI for automated real-time text toxicity classification and content filtering.
- Architected relational MySQL database schemas managing user profiles, flagged content queues, and audit logs.
- Built responsive web interfaces with complete frontend-to-backend data flow and validation.

Adaptive Traffic Signal Optimization using Computer Vision

[Python, OpenCV, YOLO, Real-Time Video Streams, Dynamic Scheduling]

- Developed real-time vehicle density detection from multi-camera feeds using OpenCV and YOLO to dynamically adjust green signal timing.
- Designed low-latency image processing pipelines reducing simulated intersection bottlenecks and waiting delays.
- Structured robust video stream preprocessing and bounding-box telemetry for dynamic signal phase allocation.

LEADERSHIP & ACADEMIC MENTORSHIP

AI Enthusiast & Peer Mentor (DSCE): Hosted hands-on mentoring sessions for engineering peers, guiding students through practical machine learning workflows, algorithms, and clean production code practices.

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